☐ Vendor Study Protocol ☐ Vendor Study Report ☑ Lpath Study Report

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In a preliminary study in a rat paclitaxel induced neuropathic pain model (1mg/kg paclitaxel intraperitoneally, every other day for 4 doses) the efficacy of the murine anti-S1P antibody and the anti-LPA antibody (504B3) was examined in comparison with their respective isotype control antibodies (LT1017 and LT1015). Antibodies were dosed intravenously (25mg/kg) one day prior to the first injection of paclitaxel and then at days 2, 5, 8, 11 and 14 after the first paclitaxel injection. Mechanical allodynia was measured by automated von Frey testing. Paclitaxel caused the development of mechanical allodynia as compared with control animals and this allodynia was significantly reduced by administration of LT1002 or 504B3 while the isotype control antibodies (LT1017 and LT1015 respectively) produced no effect on mechanical allodynia.

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Pharmacological Assessment of anti-S1P and anti-LPA Antibodies in Paclitaxel-induced Neuropathic Pain

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1 SUMMARY

In a preliminary study in a rat paclitaxel induced neuropathic pain model (1 mg/kg paclitaxel intraperitoneally, every other day for 4 doses) the efficacy of the murine anti-S1P antibody and the anti-LPA antibody (504B3) was examined in comparison with their respective isotype control antibodies (LT1017 and LT1015). Antibodies were dosed intravenously (25 mg/kg) one day prior to the first injection of paclitaxel and then at days 2, 5, 8, 11 and 14 after the first paclitaxel injection. Mechanical allodynia was measured by automated von Frey testing. Paclitaxel caused the development of mechanical allodynia as compared with control animals and this allodynia was significantly reduced by administration of LT1002 or 504B3 while the isotype control antibodies (LT1017 and LT1015 respectively) produced no effect on mechanical allodynia.

2 INTRODUCTION

The bioactive lipid lysophosphatidic acid (LPA) is an important extracellular signaling molecule in neuropathic pain. LPA and its receptors are responsible for long-lasting mechanical allodynia and thermal hyperalgesia as well as demyelination and upregulation of pain-related protein expression patterns in a variety of animal models. Evidence supports the hypothesis that antagonism of the LPA signaling pathways would benefit the treatment of neuropathic pain. We have developed the first monoclonal antibody targeting LPA (anti-LPA mAb). In preliminary studies, our anti-LPA mAb has demonstrated efficacy in attenuating pain behaviors in diabetic, chemotherapeutic, collagen-induced arthritis and sciatic nerve injury induced-neuropathic pain. Furthermore, therapy with anti-LPA Ab provides a long lasting sustained effect in accordance with the pharmacokinetic profile of the antibody. It is well accepted that the use of chemotherapy can produce neuropathic pain symptoms in patients undergoing cancer therapy. The studies described here were developed to understand if the neuropathic pain induced by the chemotherapeutic, paclitaxel, could be ameliorated by the use of an anti-LPA antibody in a rat model.

3 MATERIALS AND METHODS

3.1 Test Article and Controls

Murine anti-S1P antibody (LT1002) Lot# HR91171

Murine nonspecific IgG1 mAb control (LT1017) Lot# HR101218

Murine anti-LPA antibody (504B3) Lot# HR101201

Murine nonspecific IgG2b mAb control (LT1015) Lot# HR91184

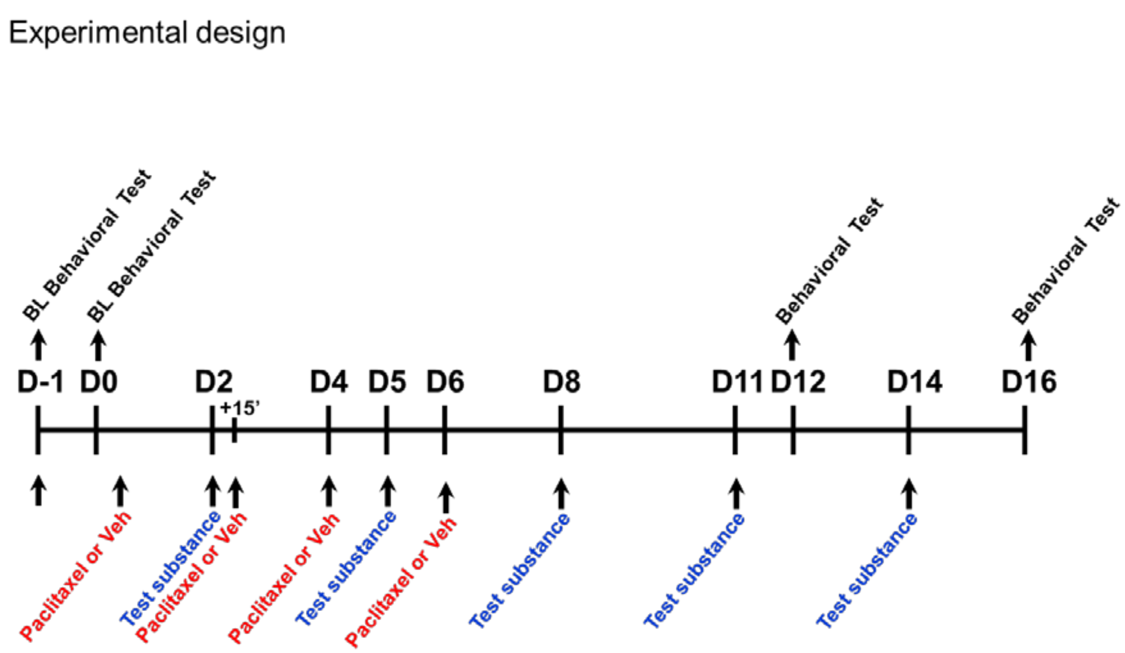
3.2 Experimental animals

Male Sprague Dawley rats (200-210g starting weight) from Harlan (Indianapolis, IN) were housed 3-4 per cage in a controlled environment (12 h light/dark cycle) with food and water available *ad libitum*. All experiments were performed in accordance with the International Association for the Study of Pain and the National Institutes of Health guidelines on laboratory animal welfare and the recommendations by Saint Louis University Institutional Animal Care and Use Committee. All experiments were conducted with the experimenters blinded to treatment conditions.

3.3 Paclitaxel-induced neuropathic pain and drug administration

The well characterized rat model developed by Bennett in which repeated intraperitoneal (i.p) injections of low doses of paclitaxel induce neuropathic pain (mechano-allodynia) with little systemic toxicity or motor impairment.[1-3] The behavioral responses last for several weeks to months, thus modeling painful neuropathies in patients. [1-3] Paclitaxel or its vehicle (Cremophor EL and 95% dehydrated ethanol in 1:1 ratio) was injected i.p in rats on four alternate days that is day (D) 0, 2, 4 and 6 at 1 mg/kg with a final cumulative dose of 4 mg/kg. [1-3] The following experimental test substances were used: LT1002, LT1017, 504B3 and LT1015; these were dissolved in saline and provided by Lpath in individual vials. Experimental test substances were given intravenously (i.v) at 25 mg/kg twice weekly as shown in Figure 1.

Figure 1 Experimental Design



Experimental test substances were given one day before (D-1) the first injection of paclitaxel, and subsequently on D2, D5, D8, D11 and D14 after the first injection of paclitaxel. Vehicle that was used to dissolve test substance (saline) was injected according to the same dosing paradigm in paclitaxel-treated group or its respective vehicle. If injections of experimental test substances coincided with the injection of paclitaxel (i.e. D2), experimental test substances were delivered 15 min before paclitaxel.

3.4 Tactile Response Threshold

Mechanical withdrawal thresholds were assessed with an electronic version of the von Frey test (dynamic plantar aesthesiometer, model 37450; Ugo Basile, Milan, Italy) on D-1 before experimental test substance injection, the day after (D0) and before the first i.p. injection of paclitaxel and subsequently on D12 and D16. To this end, each rat was placed in a Plexiglas chamber (28 x 40 x 35-cm, wire mesh floor) and allowed to acclimate for fifteen minutes. After acclimation, a servo-controlled mechanical stimulus (a pointed metallic filament) was applied to the plantar surface, which exerts a progressively increasing punctate pressure, reaching up to 50 g within 10s. The pressure evoking a clear voluntary hind-paw withdrawal response was recorded automatically and taken as the mechanical threshold index. Mechanical threshold was assessed three times at each time point to yield a mean value, which was reported as mean absolute threshold (grams, g). The development of mechano-allodynia was evidenced by a significant ($P<0.05$) reduction in mechanical mean absolute paw-withdrawal thresholds (grams, g) at forces that failed to elicit withdrawal responses before paclitaxel treatment (baseline). Paclitaxel treatments results in bilateral allodynia. [1-3] Because thresholds did not differ between left and right hind paws at any time point in any group, values from both paws were averaged for further analysis and data presentation. A total of six groups were used with n=3 rats/group.

<u>Group 1:</u> Vehicle instead of paclitaxel+saline

<u>Group 2:</u> Paclitaxel+saline

<u>Group 3:</u> Paclitaxel+ LT1002

<u>Group 4:</u> Paclitaxel+ LT1017

<u>Group 5:</u> Paclitaxel+ 504B3

<u>Group 6:</u> Paclitaxel+ LT1015

Paw withdrawal threshold (g) for groups 1-6 on (D-1) and before i.v injection of experimental test substances or their vehicle (saline) were (mean+/s.em): 43.6 ± 0.296, 43.3 ± 0.376, 43.7 ± 0.333, 42.9 ± 0.219, 42.8 ± 0.120 and 42.7 ±0.203, respectively.

3.5 Statistical Analysis.

Data are expressed as mean ± SEM for 3 animals per group and analyzed by two-tailed, two-way ANOVA with Bonferroni *post hoc* comparisons to the paclitaxel group. $*P<0.05$, $**P<0.001$ paclitaxel *vs.* vehicle group.

3.6 Results

Animal weights measured throughout the experiment demonstrated no difference across all groups (Table 1)

When compared with the vehicle treated group, administration of paclitaxel led to the development of mechano-allodynia (Figure 2 and Figure 3).

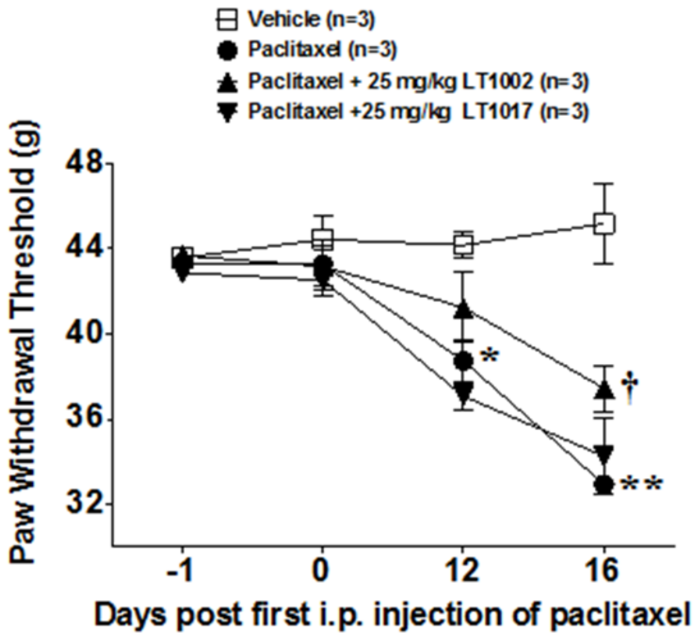
Effects of LT1002 and LT1017 on paclitaxel-induced neuropathic pain. The development of mechano-allodynia at 16 days was significantly attenuated by LT1002, but not by LT1017 (Figure 2).

Effects of 504B3 and LT1015 on paclitaxel-induced neuropathic pain. The development of mechano-allodynia at 16 days was significantly attenuated by 504B3, but not by LT1015 **(**Figure 3**).**

4 CONCLUSIONS

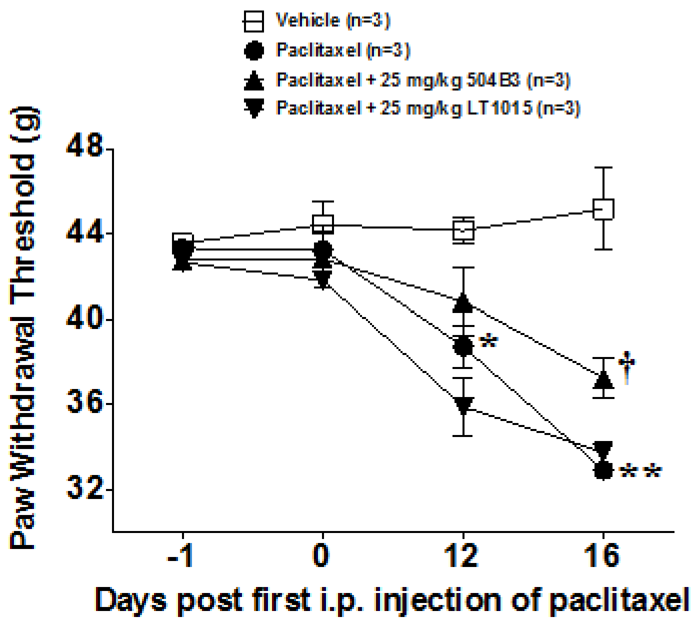
Paclitaxel caused the development of mechanical allodynia as compared with control animals and this allodynia was significantly reduced by administration of LT1002 and 504B3 while the isotype control produced no effect on mechanical allodynia.

Figure 2 Effects of LT1002 and LT1017 on paclitaxel-induced neuropathic pain.



When compared with the vehicle group (☐), administration of paclitaxel (●) led to a time-dependent development of mechano-allodynia, which was significantly attenuated at 16 days by intravenous delivery of LT1002 (▲), but not by LT1017 (▼). Results are expressed as mean ± SEM. Behavioral data for 3 animals was analyzed by two-tailed, two-way ANOVA with Bonferroni *post hoc* comparisons with the paclitaxel group where **P*<0.01 and ***P*<0.001for paclitaxel *vs* vehicle and †*P*<0.05 for paclitaxel + LT1002 *vs* paclitaxel.

Figure 3 Effects of 504B3 and LT1015 on paclitaxel-induced neuropathic pain.



When compared with the vehicle group (☐), administration of paclitaxel (●) led to a time-dependent development of mechano-allodynia, which was significantly attenuated at 16 days by intravenous delivery of 504B3 (▲), but not by LT1015 (▼). Results are expressed as mean ± SEM. Behavioral data for 3 animals was analyzed by two-tailed, two-way ANOVA with Bonferroni *post hoc* comparisons with the paclitaxel group where $^{**}P<0.01$ and $^{**}P<0.001$for paclitaxel *vs* vehicle and $\dagger P<0.05$ for paclitaxel + 504B3 *vs* paclitaxel.

Table 1 Rat Body Weights (g)

Mean±SEM					
Groups	Day -1	Day 0	Day 2	Day 4	Day 5
Vehicle+Saline	208.0±6.0	208.0±6.0	231.7±3.3	241.3±4.4	241.3±4.4
Paclitaxel+Saline	204.7±2.4	204.7±2.4	224.7±3.5	234.7±2.7	234.7±2.7
Paclitaxel +LT1002	200.7±3.5	200.7±3.5	222.7±2.9	228.0±3.5	228.0±3.5
Paclitaxel+LT1017	202.7±3.3	202.7±3.3	220.7±3.5	229.3±4.8	229.3±4.8
Paclitaxel +504B3	204.7±1.3	204.7±1.3	225.3±1.8	232.0±1.2	233.0±1.2
LT1015	205.3±1.8	205.3±1.8	225.3±0.7	233.3±1.8	233.3±1.8
Groups	Day 6	Day 8	Day 11	Day 14	Day 16
Vehicle+Saline	244.0±5.0	265.3±7.5	286.7±5.5	298.0±6.1	298.7±6.5
Paclitaxel+Saline	235.7±2.2	260.0±4.2	278.0±6.4	293.0±5.9	292.7±5.8
Paclitaxel +LT1002	228.7±4.1	247.0±3.5	266.0±2.0	276.7±2.7	281.3±3.3
Paclitaxel+LT1017	230.7±4.4	254.0±5.3	270.0±4.2	281.3±2.9	283.3±2.9
Paclitaxel +504B3	233.7±0.9	260.0±3.1	280.0±3.1	301.3±5.2	298.0±4.2
Paclitaxel +LT1015	234.0±1.2	252.7±2.7	268.0±4.2	281.3±3.3	285.3±6.2

5 LIST OF REFERENCES

1. Flatters SJ, & Bennett GJ. Studies of peripheral sensory nerves in paclitaxel-induced painful peripheral neuropathy: evidence for mitochondrial dysfunction. *Pain.* 2006;122:245-257

2. Jin HW, Flatters SJ, Xiao WH, et al. Prevention of paclitaxel-evoked painful peripheral neuropathy by acetyl-L-carnitine: effects on axonal mitochondria, sensory nerve fiber terminal arbors, and cutaneous Langerhans cells. *Exp Neurol.* 2008;210:229-237.

3. Polomano RC, Mannes AJ, Clark US, & Bennett GJ. A painful peripheral neuropathy in the rat produced by the chemotherapeutic drug, paclitaxel. *Pain*. 2001;94:293-304.